

Intelligent Blood Bank Inventory and Donor Management Platform

1. Project Vision and Stakeholders

Project Vision

To develop an intelligent, secure, and cloud-based Blood Bank Inventory and Donor Management Platform that efficiently manages blood inventory, tracks blood expiration dates, identifies eligible donors quickly, predicts future blood demand using Artificial Intelligence (AI), and improves emergency blood availability while reducing blood wastage.

Project Objectives

1. Digitally manage blood inventory.
2. Maintain secure donor records.
3. Track blood expiration dates.
4. Identify nearby eligible donors.
5. Predict future blood demand using AI.
6. Generate automated low-stock alerts.

Key Stakeholders

Stakeholder	Role
Blood Bank Administrator	Manages blood inventory and donor database
Donors	Register and donate blood
Hospitals	Request blood during emergencies
Doctors	Recommend blood requirements
Laboratory Staff	Verify blood quality and update inventory
Patients	Receive required blood
Government Health Department	Monitor blood availability
System Administrator	Maintain software, cloud database, and security

2. Epics and User Stories

Epic 1: Donor Management

User Story 1

As a donor, I want to register my details, so that I can donate blood whenever required.

User Story 2

As a donor, I want to update my profile and health information, so that my eligibility remains accurate.

User Story 3

As a blood bank administrator, I want to search donors by blood group and location, so that emergency requests can be fulfilled quickly.

Epic 2: Blood Inventory Management

User Story 4

As a blood bank administrator, I want to add newly collected blood units, so that inventory remains updated.

User Story 5

As a blood bank administrator, I want to monitor blood stock levels, so that shortages can be avoided.

Epic 3: Blood Expiration Tracking

User Story 6

As a blood bank administrator, I want to track blood expiration dates, so that expired blood is removed on time.

Epic 4: Emergency Blood Request

User Story 7

As a hospital staff member, I want to search available blood units, so that patients receive blood immediately.

User Story 8

As a hospital staff member, I want to identify nearby eligible donors, so that emergency donations can be arranged quickly.

Epic 5: AI Demand Prediction

User Story 9

As a blood bank administrator, I want AI to predict future blood demand, so that inventory planning becomes efficient.

Epic 6: Notification System

User Story 10

As a blood bank administrator, I want to receive low-stock alerts, so that I can arrange additional blood collection.

3. Acceptance Criteria

User Story 1

Given

The donor opens the registration page.

When

The donor enters valid personal and health details.

Then

The donor account should be successfully created and stored in the database.

User Story 4

Given

The administrator logs into the system.

When

New blood units are entered.

Then

The inventory should be updated immediately.

User Story 5

Given

Blood inventory exists.

When

The administrator views inventory.

Then

The dashboard should display blood-group-wise stock levels.

User Story 6**Given**

Blood units have expiry dates.

When

A blood unit is within three days of expiration.

Then

The system should generate an expiry alert.

User Story 7**Given**

Hospital staff searches for blood.

When

A blood group is entered.

Then

Available compatible blood units should be displayed.

User Story 9**Given**

Historical blood usage data exists.

When

The AI prediction module runs.

Then

The predicted blood demand should be displayed.

User Story 10

Given

Blood stock falls below the minimum threshold.

When

Inventory is updated.

Then

The system should automatically notify the administrator.

4. Product Backlog Prioritization

Priority	User Story	Business Value	Priority Level
1	Donor Registration	Very High	Must Have
2	Blood Inventory Management	Very High	Must Have
3	Blood Stock Dashboard	Very High	Must Have
4	Blood Expiration Tracking	High	Must Have
5	Hospital Blood Search	High	Must Have
6	Low Stock Alert	High	Should Have
7	Nearby Donor Search	Medium	Should Have
8	AI Demand Prediction	Medium	Could Have
9	Donor Profile Update	Medium	Could Have
10	Analytics Dashboard	Low	Future Release

5. Sprint Backlog (Sprint 1)

Sprint Duration

Selected User Stories

Story ID User Story

US-1	Donor Registration
US-4	Blood Inventory Management
US-5	Blood Stock Dashboard
US-6	Blood Expiration Tracking

Development Tasks

US-1 Donor Registration

- Design registration page
- Create donor database table
- Develop registration API
- Validate user input
- Test donor registration

US-4 Blood Inventory

- Design inventory database
- Create inventory entry form
- Develop inventory update module
- Store blood details
- Test inventory functionality

US-5 Dashboard

- Design dashboard UI
- Retrieve inventory data
- Display blood-group-wise stock
- Add search functionality
- Perform testing

US-6 Expiry Tracking

- Add expiry date field

- Develop expiry checker
- Create notification system
- Test expiry alerts

6. Effort Estimation Using Story Points

Estimation Technique

Fibonacci Sequence (1, 2, 3, 5, 8, 13)

User Story	Story Points	Justification
US-1 Donor Registration	3	Simple forms and database operations
US-4 Blood Inventory	5	Multiple CRUD operations and validation
US-5 Inventory Dashboard	5	Dashboard, filtering, and visualization
US-6 Expiry Tracking	8	Date calculations, notifications, and automation

Total Sprint Story Points = 21

Justification

- **3 Points:** Small development effort with low complexity.
- **5 Points:** Moderate functionality involving UI, backend, and database integration.
- **8 Points:** More complex feature requiring automated logic, scheduling, and notifications.

7. Sprint Goal and Expected Deliverables

Sprint Goal

Develop the core blood bank management features by enabling donor registration, blood inventory management, stock monitoring, and blood expiration tracking to establish a functional foundation for the intelligent blood bank platform.